Exercise 1 – Azure SQL Server and Database Creation

Objective

The main aim of this exercise is to understand how to create an Azure SQL Server, create a SQL database inside it, configure firewall settings, connect the database using different methods, and finally delete the resources to avoid extra cost.

Introduction

Microsoft Azure provides cloud-based services where we can create databases without installing any physical server. Azure SQL Database is a managed service which helps us store and manage structured data. In this exercise, I understood the complete process of setting up SQL resources in Azure and how they are connected step by step.

Step 1 – Creation of Resource Group

A Resource Group is like a folder in Azure that contains all related resources in one place.

1. First, we open the Azure Portal and search for *Resource Groups*.
2. Then we click on **Create** and select the subscription (Free Trial or Pay-As-You-Go).
3. We enter the Resource Group name as **GL-SQL-RG** and select the region.
4. After clicking **Review + Create**, we click **Create**.

Now the resource group is successfully created, and all further resources will be stored inside it.

Step 2 – Creation of Azure SQL Server

Azure SQL Server acts as a logical server that manages databases.

1. In the Azure Portal, we search for **SQL Servers** and click on **Create**.
2. We select the subscription and choose the resource group **GL-SQL-RG**.
3. The server name is given as **glsqlserver01**.
4. Authentication method is selected as **SQL Authentication**.
5. Admin login name is given as **mbadmin**, and a secure password is created.

After reviewing the details, we click **Create**.
Within a few minutes, the SQL Server gets deployed.

Step 3 – Creation of SQL Database

After creating the server, the next step is to create a database.

1. We search for **SQL Databases** and click on **Create**.
2. The resource group is selected as **GL-SQL-RG**.
3. The database name is entered as **SampleDB**.
4. The server selected is **glsqlserver01**.

In the additional settings, we select:
Data source → **Use existing data → Sample**

Then we click **Review + Create** and finally click **Create**.

This creates a database with sample tables and data.

At this stage, I could clearly understand the difference between a server and a database, which was confusing at first.

Step 4 – Configuring Firewall Settings

Firewall settings are important because without them we cannot access the SQL Server from our system.

1. We open the created SQL Server and go to **Firewalls and virtual networks**.
2. Then we click on **Add Client IP** and save the changes.

This step ensures that our current IP address is allowed to connect to the database securely.

Step 5 – Connecting Using Query Editor

Azure provides a built-in Query Editor to run SQL queries directly from the portal.

I opened **SampleDB** and select **Query Editor (Preview)**.
I login using:

- Username: mbadmin
- Password: (created password)

After login, we enter the following SQL query:

select * from [SalesLT].[Customer];

When we click **Run**, the customer table data is displayed on the screen.

Step 6 – Connecting Using Client Tool (Azure Data Studio)

In this step, we connect to the database using a client tool.

1. First, a Virtual Machine is created inside the same resource group.
2. Then firewall rules are updated to allow the VM's public IP address.

Inside the Virtual Machine, Azure Data Studio is installed.
A new connection is created with the following details:

- Connection type: Microsoft SQL Server
- Server name: glsqlserver01
- Authentication type: SQL Login
- Username: mbadmin
- Password: (database password)

After connecting to **SampleDB**, the same query is executed:

select * from [SalesLT].[Customer];

The results are displayed successfully.

Step 7 – Deleting the Resources

To prevent unnecessary billing, all created resources must be deleted after completion.

1. We go to **Resource Groups**, select **GL-SQL-RG**, and click on **Delete Resource Group**.
2. After confirming the name, we click **Delete**.

All resources including SQL Server, Database, and Virtual Machine are removed.

Conclusion

From this exercise, I understood how Azure SQL services are created and managed in the cloud environment. I learned how a resource group helps in organizing resources, how firewall settings are important for secure access, and how to connect a database using both Query Editor and client tools. Overall, this exercise helped me understand how databases work in cloud environment and how Azure manages everything in an organized way.